WILDY JOHANES SIMANJUNTAK

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PROFESSIONAL SUMMARY

Results-driven Software Engineer and Unity Developer with 3+ years of specialized experience in architecting high-performance interactive 3D simulations, WebXR environments, and modern web applications. Proven track record at Labtech Penta International in optimizing simulation frame rates, reducing latency, and building scalable frontend systems using Nuxt.js, Vue.js, and Tailwind CSS. Adept at bridging technical simulation architecture with seamless, user-centric interfaces.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Programming Languages:** C#, JavaScript (ES6+), TypeScript, HTML5, CSS3/SCSS
- **Frontend & Web Frameworks:** Nuxt.js, Vue.js, Tailwind CSS, Three.js, RESTful APIs
- **Game & Immersive Technologies:** Unity (2D/3D), WebXR (VR/AR), Asset Optimization, Profiling & Performance Tuning
- **Databases & Developer Tools:** MongoDB, Git, GitHub, CI/CD, VS Code, Unity Profiler
- **Core Competencies:** Interactive Simulation Architecture, LMS Integration & Maintenance, Cross-Browser Compatibility, UX Optimization

PROFESSIONAL EXPERIENCE

Labtech Penta International

Unity Developer (Contract) | Apr 2023 – Present | Batam, Riau Islands, Indonesia (On-site)

- Architected and engineered 15+ interactive 2D/3D simulations using Unity and C#, improving real-time graphic performance and boosting user engagement by 35%.
- Conducted extensive performance profiling and asset optimization (draw call reduction, texture atlas packaging, LOD management), cutting memory usage by 25% and ensuring steady 60 FPS execution.
- Integrated intuitive UI/UX workflows and WebXR capabilities to enhance simulation immersion and maintain seamless cross-platform functionality.

Labtech Penta International

Web Software Developer (Internship) | Jul 2022 – Mar 2023 | Sekupang, Riau Islands, Indonesia

- Developed and shipped responsive web modules utilizing Vue.js, Nuxt.js, and Tailwind CSS, increasing page load speed by 30% and maintaining full cross-browser accessibility.

- Maintained and optimized Learning Management Systems (LMS) serving over 1,000 active users, diagnosing technical bugs, streamlining workflows, and maintaining 99.8% system uptime.

KEY PROJECTS & SIMULATION PORTFOLIO

Interactive 3D & XR Simulations (zSpace & WebXR)

Unity Developer & Technical Simulation Architect | Labtech Penta International

- **HVAC Simulation Modules:** Engineered realistic interactive modules covering thermodynamic components and maintenance procedures, including Evaporator, Condenser, Receiver, Oil Separator, Vacuum & Evacuation, and Recovery & Recharging cycles.
- **Electrical Engineering Modules:** Developed real-time circuit logic and interactive experiment systems for Ohm's Law, Residential & Commercial Switches, and complete Residential Wiring installations.
- **Automotive & EV Technology:** Architected interactive 3D simulations detailing Electric Motorcycle Engine mechanics and Battery Management System (BMS) energy flow visualization.

EDUCATION

Politeknik Negeri Batam

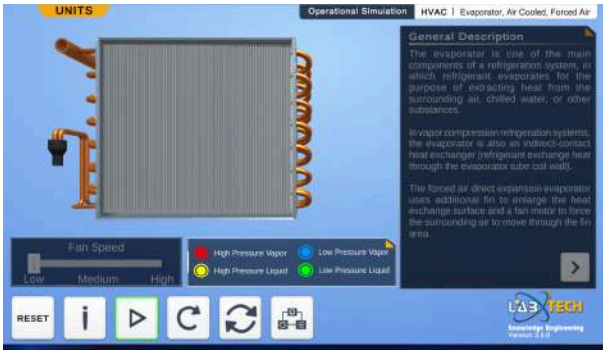
Diploma III (D3) in Informatics Engineering | GPA: 3.47/4.00

- **Relevant Coursework & Focus:** Software Engineering, Object-Oriented Programming (OOP), Data Structures & Algorithms, Web Architecture, and Computer Graphics.

LANGUAGES

- **Indonesian:** Native / Bilingual
- **English:** Professional Working Proficiency

PORTFOLIO VISUAL DOCUMENTATION (SIMULATION SCREENSHOTS)

























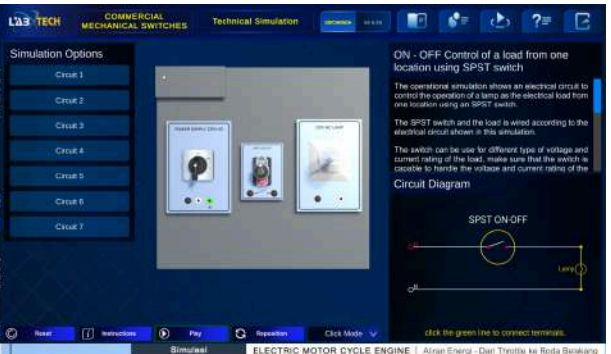

OIL SEPARATOR
Operational Simulation (Helical Oil Separator)
Helical Oil Separator
On Cycle
Display Options Panel
Inlet Connection
Outlet Connection
Case
Description
Oil separator is installed in the compressor discharge line. The oil separator is usually a vertical canister with the inlet and outlet gas connections, and an oil return port at the bottom.
The return line may be piped directly to the suction line on single compressor unit, or would be piped to an oil reservoir on multiple compressor systems.
Next
Refrigeration System
Condenser
Check Valve
Compressor
Receiver
Accumulator
Filter Drier
Oil Separator
Color Particles
High Pressure
Oil
Reset
Instructions
Pause
Reposition

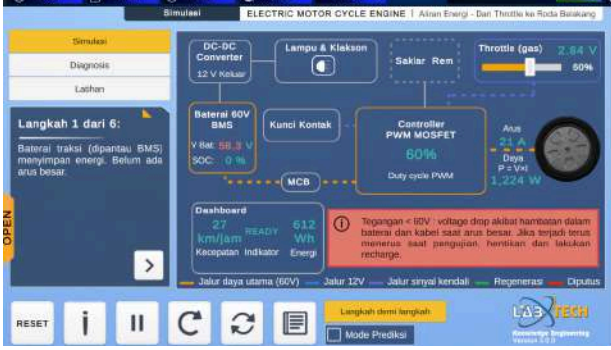

OIL SEPARATOR
Operational Simulation (Conventional Oil Separator)
Conventional Oil Separator
On Cycle
Display Options Panel
In-Out Shaping Plate
Case
Oil Return Pipe
Description
Oil separator is installed in the compressor discharge line. The oil separator is usually a vertical canister with the inlet and outlet gas connections, and an oil return port at the bottom.
The return line may be piped directly to the suction line on single compressor unit, or would be piped to an oil reservoir on multiple compressor systems.
Next
Refrigeration System
Condenser
Check Valve
Compressor
Receiver
Accumulator
Filter Drier
Oil Separator
Color of Particles
High Pressure
Oil
Reset
Instructions
Pause
Reposition


OHM'S LAW, VOLTAGE, CURRENT AND RESISTANCE
Operational Simulation
Ohm's Law
By knowing any two values of the Voltage, Current or Resistance quantities we can use Ohm's Law to find the third missing value. Ohms Law is used extensively in analyzing electronic circuits, and it is "very important to understand and accurately remember these formulas".
In a linear circuit consists of constant resistance, if the voltage is increased, the current also increases, and if the voltage is reduced, the current goes down. This also means that high voltage creates high current, and low voltage creates low current as long as the resistance remains at constant value.
Interactive Instruction
1. Press Play button to start the Operational Simulation
DC Power Supply
1000
V
mA
Reset
Instructions
Play
Reposition
Click Mode
Ohm's Law


RESIDENTIAL SWITCH
Technical Simulation
Simulation Options
Circuit 1
Circuit 2
Circuit 3
Circuit 4
Circuit 5
Control a lamp from one location
The operational simulation shows an electrical circuit to control the operation of a light from one location.
The single-pole switch and the lamp is placed in a residential building according to the electrical wiring layout used by the contractor to install the switch and light in appropriate location on the wall and ceiling.
1. Observe the control circuit and compare to the circuit diagram.
Circuit Diagrams
L
N
Reset
Instructions
Play
Reposition
Click Mode
Click the green line to connect the live/phase conductor.


RESIDENTIAL WIRING
Operational Simulation
House Wiring Diagram
Circuit 1
Circuit 2
Circuit 3
Circuit 4
Circuit 5
Circuit 6
Circuit 7
General Description
Next
Circuit 1 Guide
DIAGRAM
Wiring at the Switch Box
Reset
Instructions
Pause
Reposition
Click Mode
NEXT


COMMERCIAL MECHANICAL SWITCHES
Technical Simulation
Simulation Options
Circuit 1
Circuit 2
Circuit 3
Circuit 4
Circuit 5
Circuit 6
Circuit 7
ON - OFF Control of a load from one location using SPST switch
The operational simulation shows an electrical circuit to control the operation of a lamp as the electrical load from one location using an SPST switch.
The SPST switch and the load is wired according to the electrical circuit shown in this simulation.
The switch can be use for different type of voltage and current rating of the load, make sure that the switch is capable to handle the voltage and current rating of the
Circuit Diagram
SPST ON-OFF
Lamp
Reset
Instructions
Play
Reposition
Click Mode
click the green line to connect terminals.


Simulasi
ELECTRIC MOTOR CYCLE ENGINE | Aliran Energi - Dari Throttle ke Roda Belakang
Simulasi
Diagnosis
Latihan
Langkah 1 dari 6:
Baterai traksi (dipantau BMS) menyimpan energi. Belum ada arus besar.
OPEN
DC-DC Converter
12 V Keluar
Lampu & Klakson
Saklar Rem
Throttle (gas)
2.64 V
50%
Baterai 60V BMS
V Bat: 58.3 V
SOC: 0 %
Kunci Kontak
Controller PWM MOSFET
60%
Duty cycle PWM
Arus
21 A
Daya
P = V×I
1,224 W
MCB
Dashboard
27
km/jam
Kecepatan Indikator
READY
612
Wh
Energi
Tegangan < 60V : voltage drop akibat hambatan dalam baterai dan kabel saat arus besar. Jika terjadi terus menerus saat pengujian, hentikan dan lakukan recharge.
Jalur daya utama (60V)
Jalur 12V
Jalur sinyal kendali
Regenerasi
Diputus
RESET
Langkah demi langkah
Mode Prediksi


Simulasi
ELECTRIC MOTOR CYCLE ENGINE | Aliran Energi - Dari Throttle ke Roda Belakang
Kunci Kontak
Fungsi : Mengaktifkan sirkuit elektronik controller (power lock)
Input : Posisi kunci
Proses : Menghubungkan power lock tanpa mengalirkan arus motor
Output : Sirkuit kontrol & dashboard aktif
Jalur : Sinyal
RESET
Langkah demi langkah
Mode Prediksi